



Design and Development of a
CNN-Based Waste Classification System

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OBJECTIVES



- To design a CNN-based system that automatically classifies waste.
- To collect and preprocess a labeled dataset of waste images.
- To train and evaluate a CNN model for accurate classification.
- To enable real-time waste classification via image or camera input.





ABSTRACT



This project focuses on the design and development of a Convolutional Neural Network (CNN) based system for automated waste classification. The system is trained on a labeled dataset of waste images to classify items into categories such as plastic, paper, metal, glass, and organic waste. It uses image preprocessing, feature extraction, and deep learning techniques to achieve accurate classification. The system supports real-time image-based waste sorting, reduces manual segregation effort, and contributes to more efficient recycling and waste management.



INTRODUCTION



- Improper waste segregation is a major challenge in effective recycling and waste management.
- Traditional manual sorting is slow, labor-intensive, and prone to human error.
- The proposed system uses a Convolutional Neural Network to automatically classify waste images into predefined categories.
- It enables faster, more accurate, and scalable waste sorting for households, industries, and recycling centers.

